

Service Quality Management: A Process-control Approach

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Introduction

Services are receiving increased management attention because of the proliferation of service industries worldwide plus their direct implications for economic performance and employment. Exact numbers are elusive but the order of magnitude in the USA alone indicates that 70 per cent of GNP and 85 per cent of employment is in the service sector and growing as we go into the 1990s. International growth is virtually assured considering the emergence of the European Community and the creation of the market-driven economies of the former Communist-bloc countries. McDonald's in Moscow has been a high-visibility example of service industry transportability.

In general, society continues to move away from an agrarian, and towards an automated industrial automation, providing many people with both idle time and discretionary income. The time and income factors provide an incubator environment for service growth that can offer convenience, entertainment and general leisure activities to an increasing number of people worldwide.

The growth environment for services has been particularly strong in the USA since the 1950s. Today, many US-based service companies are the epitome in their respective industries. Disneyland, American Express, CitiCorp, American Airlines and Federal Express set the standards for quality and productivity. Internationally, SAS, Royal Dutch Shell and Rolls-Royce are known for their excellence in product/service delivery. Regardless of national origin, these companies share a common concern for their own employees and the customer when delivering their respective service bundles. A popular theory suggests that providers should go beyond satisfying the customer and consider delighting them with the quality of the service[1]. Many successful service companies have been able to achieve this level of quality consistently but all report that accomplishment requires going back to the fundamentals. While total quality management (TQM) is the overall company pursuit, getting there means paying attention to the details along the way. An excellent primer on service quality can be found in a recent edited volume summarizing an international service quality conference held in Sweden[2]. The 21 readings discuss the multidisciplinary and multinational perspectives of service quality. The process consistently includes linking marketing and operations while considering the people involved, the internal and external customers and conforming to the various requirements presented for any given service delivery.

The goal of the current study was to use some of the traditional production operations techniques associated with quality and productivity management to develop a service process control (SPC) model that would be useful in evaluating service quality and productivity issues. In order to do this, known paradigms were slightly modified to accept both definitional and terminology shifts that would allow for evaluation of a hybrid model within the general services context. Once developed, the SPC model was evaluated for conceptual validity within the context of a specific service provider. The service examined was a travel agency operation representative of a population of more than 30,000 similar providers in the USA.

A brief literature review is presented for reference purposes and additional background on service quality issues. Detailed information and statistical data on the travel agency industry are provided in the Appendix.

Background

There is a wealth of material written on manufacturing productivity and quality issues. Much of the published literature stems from work associated with Japan's industrial rebirth following the Second World War. The programmes developed that dealt with manufacturing productivity and quality issues often took on descriptive names known by their short acronyms. Total quality management (TQM), total preventative maintenance (TPM), just-in-time (JIT), statistical quality control (SQC), statistical process control (SPC) and optimized production techniques (OPT) are well-quoted examples[5].

The foundation of many of these programmes was the quantitative assessment of product characteristics in an effort to reduce the degree of variation in the final output. Control charts with upper and lower bounds of acceptable quality measurements allowed quality audits and ongoing TQM programmes to be implemented in manufacturing operations.

The literature in services quality management continues to grow and several publications warrant noting as background references: Bown *et al.*[6]; Heskett *et al.*[7]; Zeithaml *et al.*[8] and Lovelock[9]. These service texts are edited volumes including material from both academic and management perspectives. The current study utilized foundational models and conceptual philosophies from both the manufacturing and service quality literature.

As noted by Silvestro *et al.*[10], systems for measuring service quality require attention to internal, external, hard and soft measures of both tangible and intangible aspects of the service. While complex, the results of an evaluation of six service operations showed that the diversity of measurement issues may require the selective application of specific service quality concepts without the benefit of any true universal services model.

Given this caveat, the model development undertaken in the current study focused on the transaction dynamics of services or more specifically, the interaction characteristics between the provider and customer that result in a perception of quality.

Model Development

Quality is an elusive target for most services, sometimes the magic works and other times the customer is not satisfied with the outcome of the service encounter. Unfortunately, the feedback comes mainly from those experiencing extreme conditions. Measurement of service quality on a continuous scale (variable) does not appear to be a common technique. Generally, service quality is viewed more with respect to an attribute scale, that is, the service delivered is either good or bad. The problem with this approach is that the service provider management does not know the extent of the goodness or badness and therefore is often unable to address adequately the quality issue in sufficient breadth and/or depth. The marketing discipline has generally given the problem more focus than operations management during the past decade as evidenced by the number of service marketing paradigms in the literature[6,7,10,11]. The most useful model would be one that linked marketing to the operations function and as it turns out there is such a construct available that has widespread acceptance by service managers.

Model 1

The Berry *et al.*[11] service quality model considers the production function based on a customer flowing through the service operation in the following manner:

Input → Process → Output

or

Customer's: Expectation → Experience → Satisfaction

If the experience was better than the expectation, satisfaction (and quality) is high. If the experience was less than the expectation, satisfaction/quality is low. If experience equals expectation, satisfaction is average. Furthermore, marketing and advertising create the expectation, the actual service operation delivers the experience and human behaviour establishes the relationship between the two according to Lovelock[9].

The next step was to consider how satisfaction could be measured from the perspectives of both service provider and customer and the effect of focusing more on satisfying one than the other. A service provider needs to maintain some sense of proportion in this regard in order to remain financially viable.

Model 2

A second model considered uses the concepts of efficiency and effectiveness from the perspective of productivity[12]. Given the basic definition of productivity as output over input, the model developed a case for equating output to effectiveness and input to efficiency. Tests demonstrated that the service provider was concerned with efficiency and the customer focused on effectiveness. Successful service operations must consider both of these factors in order to continue to deliver a quality experience to the customer and sustain a profitable

operation. In the terminology of Model 1, both parties to the transaction must be “satisfied” with the experience. Davis has explored the concepts of effectiveness and efficiency from the specific orientation of waiting time and cost of service[13]. A mathematical formulation was tested in a fast food operation confirming the basic relationship as proposed in Model 2.

The homeostatic model as shown in Figure 1 requires that the service transaction process constantly move towards a balance position to deliver high satisfaction (quality) to the provider (efficiency) and to the customer (effectiveness).

The key objective of the service provider is to stay close to the diagonal while pursuing activities that will evidence efficiency gains (usually cost reductions) and raise effectiveness (usually more customer services). The quality of these efforts can be measured from the perspective of the satisfaction achieved as a provider moves up the diagonal. Once a balance position is achieved, dramatic changes along a single axis could disrupt the perceived quality from either the provider or customer viewpoint. The third step in model development was to consider both provider and customer perceptiveness within the framework of a quality-control chart.

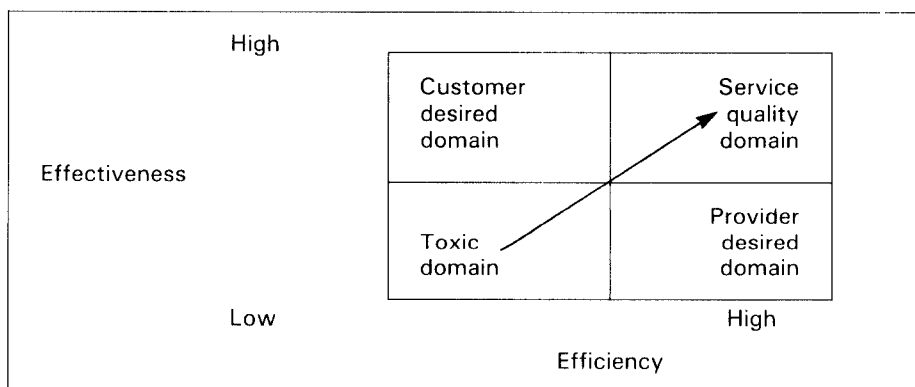


Figure 1.
The Homeostatic
Model

Assumptions

The final premise used in developing a model was that service deliverables could be considered within the context of an *R*-type control chart. The typical chart has three features, a mean value of some designated parameter, an upper acceptable limit (UCL) and a lower acceptable limit (LCL) [5]. The limits would be determined from a statistical analysis of the product and process and generally expressed as number of standard deviations from the mean. For instance, a chart based on three standard deviations or three sigma would approximate to a quality level of 95 per cent, a typical quality level across US industry. Some companies have pursued higher quality levels. Motorola[14] has established a company standard based on six sigma or more than 99.9 per cent. However, recent indications are that in certain industries, this level of quality may hinder tactical reaction time to new technology introduction.

Graphically, an *R*-chart generally appears as follows:

UCL
Mean
LCL

In actual use, repetitive samples would be taken according to a predetermined plan and have their measured values recorded on the chart. If the values should exceed either the UCL or LCL, the process could be considered to be out of control and the resulting quality of the measured tangible product to be highly suspect. However, if one could establish boundaries for a service bundle based on this concept, management could routinely evaluate the quality of the service being delivered by checking the process characteristics.

Service Process Control Model

The two models described above are now superimposed onto a control chart construct with the following characteristics:

- (1) The mean measured value of the “product bundle” is the market value of the service being delivered.
- (2) The UCL can be considered to be a measure of efficiency or the resource costs to deliver the service. Above a certain level, the provider cannot expend the resources based on the predetermined market value of the deliverable. If the service is costing too much to produce, the process is out of control.
- (3) The LCL can be considered to be a measure of effectiveness or the quality level of the service acceptable to the customer. Below a certain level, the customer will choose not to accept the service and take his/her business elsewhere. If the service is not effective, the process is out of control and figuratively beyond the hypothesized lower control limit or acceptance threshold.

The SPC model can be expressed on an *R*-chart basis as shown:

UCL → Efficiency
Market value
LCL → Effectiveness

Any given service could establish an objective to take aim at considering the feasibility of increasing both efficiency and/or effectiveness. The goal would equate to becoming more productive through managing operating costs towards a lower level while maintaining or increasing customer satisfaction. Once these ideas are accepted, management needs to establish a mechanism for evaluation. The action taken must be reasonably controlled in order to determine the direction and magnitude of the resulting change in the process and the potential equilibrium shift.

The Intervention

The designated intervention was based on the work done by Fitzsimmons and Sullivan[15] wherein three methods are proposed to increase service productivity; focusing on technology and people. First, use existing technology from manufacturing and transfer to a service operation. This idea can be seen in the production-line approach to making hamburgers at McDonald's[16]. Second, use computer/automation to standardize the service being delivered. Third, increase the actual customer participation in the service delivery process. These last two efforts can be seen in the banking industry with the use of automated teller machines that automate the teller function and require the customer to participate directly in securing the deliverable service — generally cash withdrawal from the account.

The links between productivity and quality have been well established and reported in the literature[6,7,8]. Given these associations, the SPC mode. was then tested empirically in a field setting. Readers are encouraged to see the Appendix for detailed descriptions of both the travel agency population and the sample used in the test.

Model Evaluation

The productivity enhancement concepts were framed into the travel agency operations terminology. Efficiency was generally defined as the number of customers handled by each agent and indirectly, the amount of revenue generated. The SABRE CRS offered a productivity management package that measured the number of file/record accesses along with duration of contact used in booking a travel reservation. Effectiveness was considered as having satisfied customers as a function of the number of customer complaints, complaint resolution and the amount of repeat business generated. The agency challenge, then, was to handle more customers with a minimum expenditure of resource allocation and secure additional repeat business from the existing customer base.

Applying the productivity enhancement techniques within the agency framework required assessing the current issues and problems. The quality “audit” was done using a focus planning workshop with the front-line agents, structured interviews with both employees and selected customers and mail questionnaires to a sample of the customer base. Evaluation of the data collected indicated six key issues that could be divided equally between the efficiency (FIC) and effectiveness (FEC) categories:

- (1) CRS terminal time to book reservations (FIC)
- (2) Telephone time with customer to book reservations (FIC)
- (3) Total number of commissionable bookings (FIC)
- (4) Number of interrupted transaction sequences (FEC)
- (5) Number of mistakes/complaints on bookings (FEC)
- (6) Ability to handle enquiries during off-hours (FEC).

The next step was the intervention or the selected application of the productivity enhancement tools. Several automation concepts were utilized. First, a new SABRE CRS was acquired, with 386-based PCs replacing the older dedicated terminals. The new terminal offered the agent increased transaction speed, a "windows" approach to make inter-record checking easier, and broader functions including a word-processing package built into the basic software. Next, management added a 24-hour telephone system offering the customers a pre-recorded message identifying the agency's hours plus any weekly "specials" that were available. In order to give special attention to the cruise customers, a dedicated telephone line and number were added to the office rotary for direct access. The same machine was used to answer any daytime telephone call that could not be easily picked up by the busy agents. This equipment allowed phone-ins to be serviced in a token fashion while keeping the agents relatively uninterrupted when actively serving a user. A daytime message was utilized that explained the situation to the phone-in customer and requested that they leave a message rather than wait on the line for the next available agent. Agents were asked to keep daily transaction logs for the test period that were matched to actual record access counts on the CRS system. The existing management reports captured the number of customer and provider complaints plus the resolution factors. The monthly financial ledgers were the source of actual commissionable bookings.

Discussion of Results

Measurement of the results was both qualitative and quantitative but generally suggested improved performance in both efficiency and effectiveness. Within the *R*-chart terminology, the efficiency was improved by reducing resource cost allocation per booking dollar and the UCL was lowered. Likewise, effectiveness was improved by increasing customer satisfaction and the LCL was raised. In typical SQC control chart management, this type of movement of the UCL and LCL references indicates improved quality performance (increasing sigma).

The initial quantitative performance-related results for the test period are presented for reference:

- (1) The new CRS required an estimated 11 per cent less time to book a given reservation; this time may be further improved when the agents are fully trained and familiar with the CRS operation.
- (2) Telephone time with customers was reduced by an estimated 12 per cent to handle a transaction; however this may be closely linked with the CRS performance.
- (3) The total number of bookings for the test period exceeded the forecasted levels by approximately 15 per cent to the prior months' moving average and the prior year's comparison.
- (4) The answering system reduced the number of interruptions by at least 50 per cent during daily operating hours.

- (5) The number of mistakes and complaints was somewhat reduced but actual improvement was difficult to isolate because the agency had always been very attentive to this factor previously.
- (6) The answering system generally improved after-hours telephone handling as indicated by the number of messages taken during the test period. Actual improvement is difficult to assess because of the unknown number of calls previously.

Several qualitative performance issues also emerged from post-test interviews with employees. Agents had additional “free” time to sell to new or to revisit old customers. Customers expressed appreciation for the answering system rather than being put on hold. Both customers and employees felt less pressured due to knowing that an incoming telephone call would be handled in a predetermined manner and appropriate follow-up could be done when there was available time. Generally, the agents believed themselves to be more productive and customers indicated higher levels of satisfaction following the intervention described.

The increase in quality was confirmed, in Model 1 terminology, by having the overall customer and provider employee experiences exceed existing expectations. The quality shift from a Model 2 perspective was moving upwards along the diagonal by increasing customer effectiveness while decreasing costs per transaction. The movement of the selected measures on the service process control model chart evidenced higher levels of quality and productivity as shown by increases in efficiency and effectiveness.

Conclusions

The research confirmed that managing service quality can be addressed through the use of concepts derived from manufacturing and from existing service models. The limitation of the approach presented lies in the requirement to identify measurable characteristics of both effectiveness and efficiency in any given service plus the feasibility of applying some intervention and determining the effect on service quality. Extensions of this research into other service operations may be feasible, however attention must be given to proper preparation of the measurement parameters used for quality assessment and these factors will generally not be consistent across service industries.

Many service providers have utilized a process control approach to increasing quality or productivity without specifically identifying the intervention in the context of service quality charts. Many banks installed automated teller machines (ATMs) in order to offer more customer convenience (effectiveness) while reducing transaction costs (efficiency). Self-service gas stations have been able to cut costs by reducing labour while giving the customer the increased satisfaction of pumping their own fuel (no more spills or overfilling). Disneyland has changed from having a ticket booklet of individual ride coupons to having a single charge for admittance that includes unlimited rides. Paper costs have been reduced and the customers' perception of the service experienced has been increased. Other examples exist across various service industries and all point to management being concerned enough to pursue change in the existing process.

Management Implications

The primary requirement for any manager planning to use the approach to service quality management proposed herein would be to consider the project in a systematic fashion. Common steps might be taken across virtually any service operation:

- (1) Determine what your customer values in the service transaction — this could be price, convenience, speed, personal interaction or other factors indicating effectiveness.
- (2) Determine the make-up of the transaction-cost structure from the provider perspective — what is the actual cost of an average transaction, the efficiency factor.
- (3) Establish a mechanism that can dynamically capture the effect of shifts in the effectiveness and efficiency components of the transaction. Construct a quality control chart.
- (4) Identify a feasible intervention that will cause potential shifts in the effectiveness/efficiency balance position including estimates of directorial and magnitude results.
- (5) Use a controlled intervention approach, measure the impacts on the chart and determine the follow-up actions necessary either to correct a poor result or enhance a positive shift on the chart.

The steps presented are not a panacea but only a strategic way of looking at how to do business better — or can we “delight” the customer with our service quality?

Future research should pursue testing of the proposed SPC approach in other service applications. Certainly the degree of actual and potential automation and customer involvement will be a limiting factor in some services. Whatever limitations may prevail, most services should be able to address quality opportunities from the customer-effectiveness/provider-efficiency structure on a general basis. As additional research can isolate both unique and/or common factors across services for evaluation, service management can continue to evolve as a separate disciplinary area of interest and importance.

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Appendix. The Travel Industry

The most recent industry survey of the travel business identifies the service catering to both leisure and business travellers in nearly equal proportions[3]. Transport deregulation, and specifically airline deregulation in 1978, has changed the framework of the industry during the past decade. Agency booking growth has grown from \$20 to \$80 billion over ten years. The number of agencies has grown from 14,800 to more than 30,000 in the same timeframe. Two-thirds of these agencies have an average booking revenue of less than \$2 million. Cruise sales were 15 per cent of agency sales (1989), followed by hotels (11 per cent), car rentals (8 per cent), rail (4 per cent), and other (4 per cent). Air transport continues to dominate agency business at 58 per cent of revenues. The industry expects cruise business to continue to rise during the 1990s. Domestic travel accounts for 70 per cent of agency sales today. Average agency revenue was \$2.6 million in 1989 and the typical agency employed 6.3 people. Nearly 68 per cent of agencies are single location, 19 per cent are branches and 13 per cent are home-office locations. Most (86 per cent) are not affiliated with any association for corporate travel and fewer (61 per cent) are not affiliated for vacation travel.

Technology has played a vital role in the expansion of the travel agency industry. The most significant factor is the widespread proliferation of computerized reservation systems (CRS). There are several systems on the market, each identified with a major trunk airline. The SABRE system, offered by American Airlines, commands more than 50 per cent of the market with 15,500 agency customers. Others include APOLLO (United), SYSTEM ONE (Texas Air/EOS), WORLDSPAN (Delta, TWA, NW) plus worldwide, several smaller systems are emerging. A travel agency with a CRS terminal can find the lowest fare, best schedule, book and confirm all air and ground services and even issue a boarding pass along with the ticket within a matter of minutes. The larger CRS providers offer accounting packages, frequent flyer tracking and a host of attendant software to help manage agency operations[3].

The second area of technology implementation is in the area of electronic funds transfer. The industry clearing house, Airline Remittance Corporation (ARC) clears all accounts both for carriers and agencies weekly. Should funds not be available, penalties and reserve bonds can be invoked that deplete the already small margin for the agencies. The avalanche of special fares, restricted tickets and changing fare structures has made daily agency operations an administrative nightmare for management surrounding the quality and productivity issues[3].

The single sample that was used to evaluate the proposed model has been more fully explored in a case study[4]. The agency is a typical small operation having been in business for 12 years and experienced steady growth in booking revenues to \$1.7 million in 1989. The single office

is in a downtown location and has five full-time employees plus two part-timers. Their market position is currently third in the city with just over 17 per cent of the total travel agency business. Management believed that profits were being eroded by changes in the agency mechanism. Agencies operate on a steady 10 per cent margin, with cruise margins being somewhat more (ranging from 12 to 17 per cent), but customers are demanding more service and the travel providers (airlines, cruise lines, hotels, etc.) are requiring the agencies to do more for the commission. Management was proud that all agents were certified travel consultants (CTCs), identifying them as having attended special courses and being "certified" as professionals. A recent study by a small business industry (SBI) project group from the local university indicated that the agency's financial performance was better than the industry ratios but that growth was stagnant.

The agency owner co-operated fully with this study because she was convinced that both quality and productivity could be enhanced in day-to-day operations. She participated in establishing the baseline operational parameters and selecting interventions to be evaluated within the SPC model construct.